SC4063 PART 2 : DELIVERABLE 3

INTRODUCTION

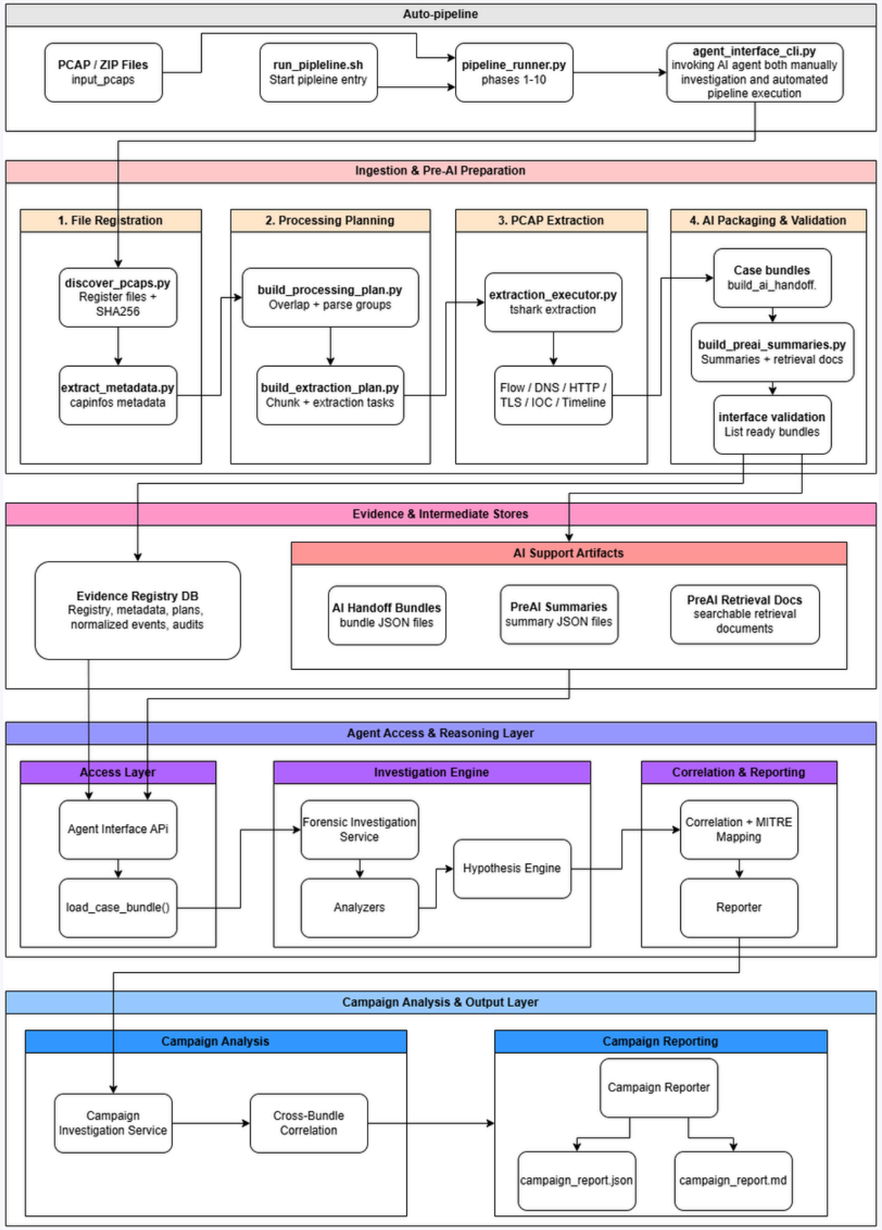
- Agent-based architecture for automated PCAP analysis
- Multi-stage pipeline : ingestion, extraction, correlation, reasoning
- Integrates protocol analyzers (DNS, HTTP, TLS, beaconing, IOC)
- Produces structured outputs and campaign-level insights

1. AGENT ARCHITECTURE

COMPONENTS + DATA FLOW

- PCAP files are ingested and preprocessed
- Ingestion pipeline extracts metadata and prepares data
- Agent acts as a central controller of the system
- Analysis modules extract protocol-level evidence
- Results are correlated and converted into insights

COMPONENTS + DATA FLOW



TOOLS AVAILABLE TO THE AGENT

- DNS Analyzer : identifies suspicious domains and queries
- HTTP Analyzer : inspects web traffic and requests
- TLS Analyzer : analyzes encrypted connections
- Beaconing Detector : detects periodic communication patterns
- Threat Intelligence Module : enriches data with known indicators

DECISION-MAKING LOGIC

- Agent dynamically selects analysis modules
- Decisions are based on extracted evidence
- Results from one module guide the next step
- Correlation engine links related events
- Hypothesis engine generates possible attack scenarios

2. DEMONSTRATION

3. KEY CHALLENGES

TECHNICAL CHALLENGES

- Large PCAP processing is slow and resource-heavy
- Correlation is hard across multiple extractor outputs (time, host, protocol)
- Uses normalized Zeek/Suricata-style events, but relies on fixed tools (capinfos, tshark)

FALSE POSITIVES / MISSED DETECTIONS

- Legitimate traffic (CDNs, updates, APIs) may appear suspicious
- Stealthy attacks can evade behavioural detection
- TLS encryption hides payloads - analysis limited to metadata (SNI, certs, JA3)

TOOL LIMITATIONS

- Extractors provide evidence, not conclusions
- Conclusions require combining multiple analyzers
- Detection depends on PCAP quality (missing / partial data reduces accuracy)

SCALING / PERFORMANCE ISSUES

- Large PCAPs increase processing time and storage
- Sequential pipeline adds latency
- Deeper analysis increases compute cost
- Trade-off : speed vs coverage vs confidence

4. GUARDRAILS AND SAFETY CONTROLS

PREVENTING HALLUCINATION

- Evidence-driven reasoning only
- No direct access to raw PCAPs
- Works via structured data layer
- Unsupported hypotheses → **flagged / downgraded**

AGENT CONSTRAINTS

- Fixed investigation workflow
  - (Summary → DNS/HTTP/TLS → IOC → MITRE → Findings)
- Uses **AgentInterface + predefined analyzers only**
- No direct DB queries or raw PCAP parsing
- External tools restricted (e.g., tashark, capinfos)

VALIDATION BEFORE CONCLUSIONS

- Multi-factor scoring:
  - Evidence strength
  - Number of sources
  - Source diversity
- Cross-signal correlation:
  - DNS + HTTP + TLS + Beaconing + IOC
- Minimum evidence threshold required
- Confidence level determines reportability
- Full **timeline** + **provenance tracking**

HUMAN-IN-THE-LOOP

- High-risk / low-confidence → **analyst review**
- Reports include:
  - Confidence level
  - Evidence items
  - Risk (FP /FN)
  - Limitations
- Agent = **Decision support**, not replacement

5. COST AND EFFICIENCY

COMPUTATION OF COST PER RUN (ESTIMATED OR MEASURED)

- Local execution (CPU-based)
- Assumed rate: $0.05 per CPU-hour
- Total runtime: 94.1 minutes
- Estimated cost: ~ $0.078 per run

Key Point

- No cloud LLM cost (local pipeline)
- Cost mainly comes from CPU processing of PCAPs

- Note: All runtime and cost values in this table are based on the saved pipeline_run_metrics.json file from a successful full pipeline run. Additional task-level efficiency statistics are derived from evidence_registry.db using injestion/view_run_metrics.py.

TIME TO COMPLETE ANALYSIS

Stage	Runtime	Notes
Full Pipeline	94.1 min	End-to-end
Evidence Extraction	93.9 min	Main bottleneck
Agent Investigation	0.49 sec	Near real-time
Campaign Analysis	0.46 sec	Very fast
Final Synthesis	0.06 sec	Negligible

- Note: All runtime and cost values in this table are based on the saved pipeline_run_metrics.json file from a successful full pipeline run. Additional task-level efficiency statistics are derived from evidence_registry.db using injestion/view_run_metrics.py.

TRADE-OFFS BETWEEN SPEED, ACCURACY AND COST

- Note: All runtime and cost values in this table are based on the saved pipeline_run_metrics.json file from a successful full pipeline run. Additional task-level efficiency statistics are derived from evidence_registry.db using injestion/view_run_metrics.py.

6.CODE ANALYSIS

Agentic AI systems can hallucinate - reporting IPs, domains, or ports that never existed in the data

CAN WE TRUST AI FINDINGS?

Agent Finding

Finding: Suspicious TLS Session

Evidence: 10.128.239.57 → 88.214.25.115:34985

Confidence: 0.73 (MEDIUM)

MITRE: T1573, T1071

Does it match?

Raw PCAP Events

tls event

src: 10.128.239.57
dst: 88.214.25.115
dst_port: 34985
sni: (null)

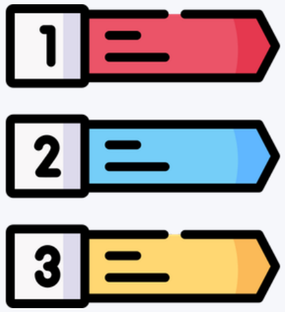
flow event

10.128.239.57:3389 → 88.214.25.115:34985

Grounding check: every evidence is independently verified against indexed raw PCAP files

VALIDATION METHODOLOGY

Index Raw Events

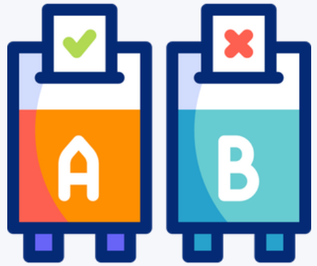


- Parse each bundle's ai_handoff_bundles JSON
- Build lookup sets: DNS domains, TLS (src, dst, port) tuples, flow pairs and all IP addresse

Check Each Evidence Item

For every evidence item in every finding, run source-specific checks (DNS, TLS, beaconing against the index)

Record Pass/Fail



- Each item is marked Grounded or Ungrounded.
- Failures log the exact reason for audit purposes.

VALIDATION RESULTS

52
Total Bundles Processed

34
Bundles with Findings

361
Evidence Items Checked

100%
Grounding Rate

Bundles PASS - all evidence grounded	34/34	All passing
Bundles FAIL- all ungrounded evidence	0/34	None
Evidence items grounded	361/361	100.0%
Evidence items ungrounded (hallucinated)	0/361	0%
Overall grounding rate	NIL	100.0%

FINDINGS BREAKDOWN

Suspicious TLS Session

MITRE: T1573, T1071

Evidence items: 289

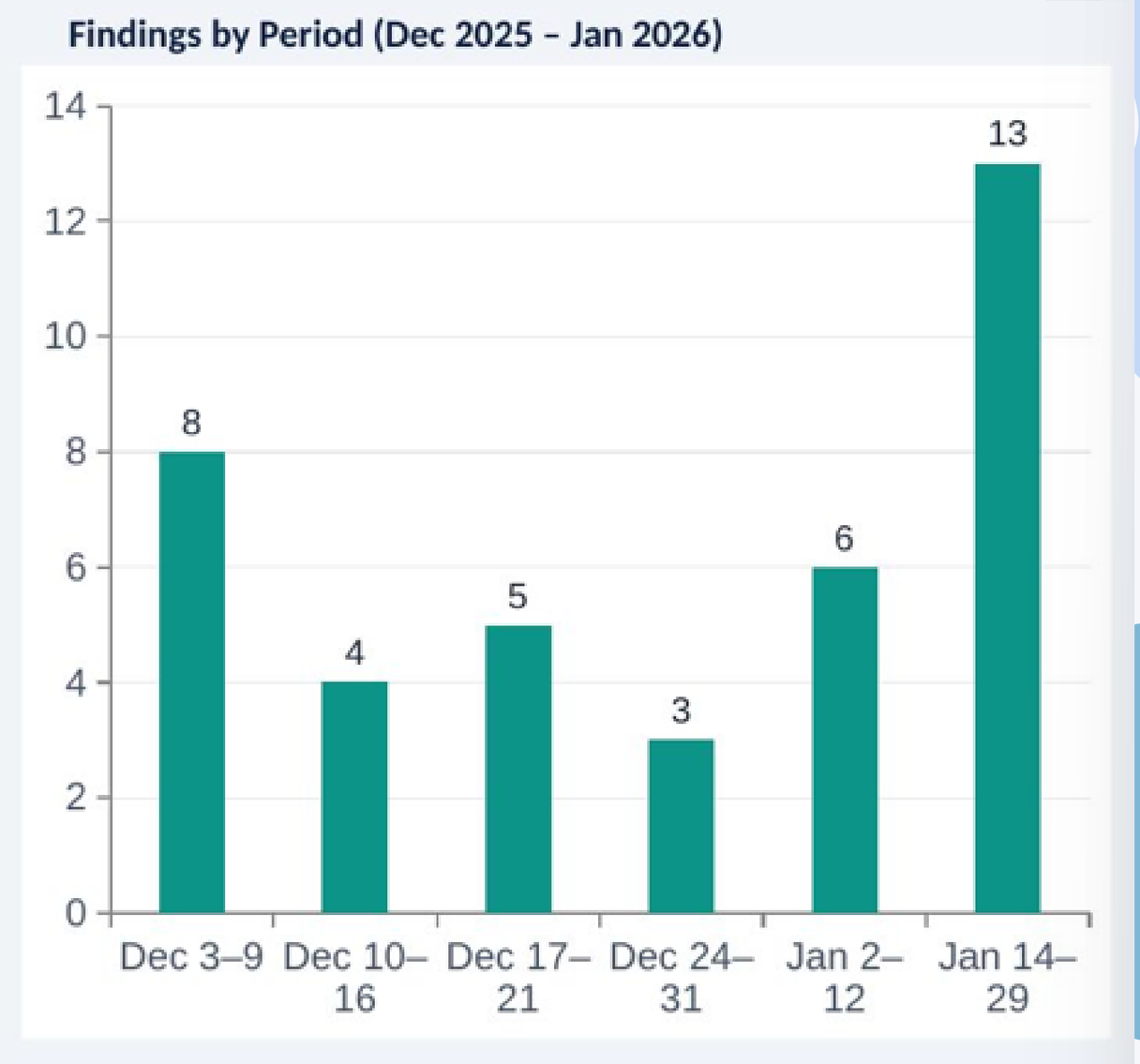
All grounded: 100%

Suspicious DNS Session

MITRE: T1071.004

Evidence items: 72

All grounded: 100%



SUSPICIOUS TLS SESSIONS – GROUNDING VERIFIED

Pattern Detected

- 39 unique external IPs connected to internal host 10.128.239.57 on port 3389 (RDP)
- All TLS sessions lacked SNI (hallmark of automated or malicious tooling)
- TLS also observed on non-stamdard ports alongside RDP
- Activity persisted from Dec 3, 2025 through Jan 29, 2026 (nearly 2 months)

Top Repeat External IPs

179.60.146.36	**20x**	**10.128.239.57: 3389**
194.165.17.11	19x	10.128.239.57: 3389
179.60.146.33	17x	10.128.239.57: 3389
141.98.11.114	12x	10.128.239.57: 3389
195.211.190.189	10x	10.128.239.57:3389

CONCLUSION

100%
Zero hallucinations

- 361 of 361 evidence items verified against raw PCAP data.
- The agent cited no IPs, domains, or ports absent from the original capture.

RDP
Real Threat Signal

- 39 unique external IPs persistently targeting 10.128.239.57:3389 across 34 days
- a grounded, unverifiable finding surfaced automatically

Safe
Comservative Design

- DNS findings carry explicit false-positive warnings.
- Confidence thresholds and minimum evidence rules prevent over-reporting.

CONCLUSION

- End-to-end agent-driven PCAP analysis pipeline
- Evidence-based correlation across multiple analyzers
- Structured outputs with confidence and traceability
- Limitations from data quality, tools, and detection constraints
- Supports scalable, explainable network forensics

THANK YOU